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On the Exterior Calculus of Invariant Theory

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1. INTRODUCTION

The full extent of the tragedy of Hermann Grassmann, which has been unfolding since his death, by a succession of misadventures and misunderstandings of his work unique in the history of modern mathematics, is yet to be fully realized. It began with a deliberate ambiguity on the part of the author of the “Ausdehnungslehre,” whereby one and the same notation was chosen to designate two different algebraic operations, originally known as the progressive and the regressive product. This choice was not an oversight on Grassmann’s part, as we shall see, but was rather dictated by the author’s deep understanding of the philosophical reach of his discovery, an understanding which lay well beyond the powers of the

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ordinary mathematician of the time. Grassmann's frequent excursions into metamathematical considerations were indeed mingled with mathematical discoveries which a few courageous readers did not hesitate in qualifying as awesome. On the whole, however, the style of the "Ausdehnungslehre" clashed with the ideal of axiomatic presentation which was triumphantly emerging at the time and was therefore regarded with suspicion.

There is strong evidence to indicate that most mainstream algebraists of the time, such as Gordan, Capelli, Hermite, Cayley, Sylvester and even Hilbert, while paying lip service to Grassmann in an occasional footnote, did not realize the sweeping extent of Grassmann's discovery and its relevance in invariant theory.

The epigons of invariant theory in this century, such mathematicians as Turnbull, Aitken, Alfred Young, Littlewood and even Hermann Weyl, perpetuated the same sin of omission, and one finds in these authors' work scattered rediscoveries and partial glimpses of ideas that could have been made to bloom, had the authors used even only the notation of exterior algebra.

There were, to be sure, some exceptions. To the best of our knowledge of published work, the first mathematicians to understand, albeit imperfectly, the program of the *Ausdehnungslehre* were Clifford and Schröder, first, and later A. N. Whitehead, Elie Cartan, and best of all Giuseppe Peano. The present work is meant to develop the program undertaken by Grassmann and pursued by these authors.

As often happens in mathematics, the crucial step is the choice of a notation which is fully in line with the concepts to be understood, a task that is nowadays made more difficult by the currency of other notations of the day, developed for specific but limited objectives.

It was Schröder, in an appendix to his "Algebra der Logik," who first stressed the analogy between the algebra of progressive and regressive products, and the algebra of sets with union and intersection. Alfred North Whitehead further preached this analogy by presenting in parallel chapters, in his "Universal Algebra," first the theory of Boolean algebra in the cup/cap notation invented by Peano, and then the "algebra of extension." Whitehead, however, did not take the decisive step of clearly distinguishing the two kinds of products, let alone axiomatizing them. Nor could he have done so, since the very concept of a ring was to emerge afterwards. His work is largely a rewrite, somewhat clearer to be sure, of some of Grassmann's work. Even Whitehead's considerable stature in the English-speaking intellectual world, however, was insufficient to sway the mathematicians of his time in the direction pointed by Grassmann.

Elie Cartan took a considerable step forward, but in the process he paid a price of omission from which we are yet to recover. He realized the usefulness of the notion of exterior algebra in his theory of integral

invariants, which was later to turn into the potent theory of differential forms. By this somewhat devious route he succeeded in convincing the mathematical public that exterior algebra is an indispensable tool of a mathematician, a conviction that was eventually made into accepted dogma with the publication of the third chapter of Bourbaki's "Algèbre." Unfortunately, with the rise of functional analysis, another dogma was making headway at the time, namely, the distinction between a vector space V and its dual V^* , and the pairing of the two viewed as a bilinear form. Elie Cartan found the regressive product to be superfluous and awkward. By vector space duality, a pairing of the two exterior algebras of V and V^* could easily be made with only one kind of product, the one that came to be called the wedge. The regressive product was quietly shelved, and was replaced with the unwieldy notation of interior products, whereby an extensor x in $G(V)$ is viewed as an operator acting on an extensor u in $G(V^*)$, an action designated by the symbol $x \lrcorner u$. Thus, if V and V^* are kept as different spaces interacting by interior products, the symmetry between x and u is lost. Worse, computations with interior products are openly acknowledged to be cumbersome and unintuitive, even in such a smooth presentation as the second edition of Bourbaki's "Algèbre Multilinéaire." Worse yet, the catchy analogy between interior products in exterior algebra and the analogous products in the symmetric algebra, provided by the common presentation of both in the language of Hopf algebras, further obscures the basic fact that the exterior algebra is a bird of a different feather. The interior product notation works wonders in the symmetric case, but, in the antisymmetric case, it fails to tell the full story. If one insists in keeping interior products, one is sooner or later faced with the symmetry of exterior algebra as a Hopf algebra. Unlike the symmetric algebra, the exterior algebra of a finite-dimensional vector space is isomorphic to its dual as a Hopf algebra. This isomorphism is ill-matched with the asymmetry of interior products. It can be expressed by remarking that interior products in exterior algebra satisfy additional identities not shared by interior products of the symmetric algebra. At bottom, the key fact is that the determinant of the product of two matrices equals the product of the determinants, while no such property holds for the permanent. The lack of symmetry in such a presentation of exterior algebra becomes glaring in its geometric interpretation. Grassmann's idea was to develop a calculus for the join and meet of linear varieties in projective space, a calculus that is actually realized by the progressive and regressive products. It has been amply demonstrated that this calculus furnishes the definitive notation for such computations. It would, however, be capricious to limit such a calculus to a single operation, just as capricious as limiting the algebra of sets to the single operation of union. Furthermore, the dual space V^* of a vector space V plays no role in such a calculus: a hyperplane

is an object living in V , and its identification with a linear functional is a step backwards in clarity. What is needed is an extension of the exterior algebra of V which introduces the second operation—the meet, as we call it—without appealing to vector space duality, and in a notation that is likely to provide the utmost transparency in simultaneously computing with both operations. This is, in brief, the objective of the present work. We submit that the notion of a Peano space, and its concomitant double algebra, is the natural algebraic structure for the study of invariant theory under the special linear group.

To motivate the notion of a Peano space, consider the analogous situations for the orthogonal and symmetric groups. We have:

(1) A *Hilbert space* is a vector space V endowed with a symmetric bilinear form (x, y) for $x, y \in V$, called the *inner product*. The orthogonal group is the group of all linear transformation leaving the inner product invariant. Invariants under the orthogonal group are expressed as polynomials in inner products. Computations in Hilbert space are actually carried out in an algebra associated with the inner product, called the Clifford algebra.

(2) A *symplectic space* is a vector space V endowed with an antisymmetric bilinear form $[x, y]$, defined for $x, y \in V$. The symplectic group is the group of all linear transformations leaving the symplectic form invariant. Invariants under the symplectic group are expressed as polynomials in the symplectic form. Computations in a symplectic space are carried out in an algebra naturally associated with the symplectic form, called the Weyl algebra.

One is led to ask whether a similar statement holds for the linear group. We are thus led to the following definition.

(3) A *Peano space* is a vector space V of dimension n , endowed with an antisymmetric non-degenerate multilinear form $[x_1, \dots, x_n]$, defined for $x_i \in V$, called the *bracket*. The special linear group is the group of all linear transformations preserving the bracket. Invariants under the special linear group are expressed as polynomials in brackets (first fundamental theorem of invariant theory). Computations in a Peano space are naturally carried out in an algebra associated with the bracket, which we propose to call the *double algebra*.

The main difference between (1) and (3) is that the double algebra has two operations instead of one; it turns out that there is even a third operation, the Hodge star operator, which is the vector space analog of complementation in Boolean algebra.

Not unexpectedly, the operations are called join and meet. The double algebra of a Peano space, together with the Hodge star operator, fulfills

Grassmann's program of providing an algebraic underpinning to computations with linear varieties in a vector space, in close parallel to the operations of union, intersection and complement in Boolean algebra.

The bulk of this work consists in a study of the identities holding for meets and joins of extensors (read: decomposable antisymmetric tensors), and in developing a notation which is suited to such computations. The deeper such identity, and, we believe, a novel one, is the last alternative law (Theorem 7.10). This identity is crucial in obtaining the various straightening algorithms of invariant theory, in fact it leads to several other canonical forms for expressions in the double algebra.

The divergence between the present notation and other alternative approaches that rely to varying degrees on the underlying Hopf algebra structure can be clearly seen in Proposition 4.2, where it is shown that the meet of two (or more) extensors can be computed by two radically different algorithms. This is the point at which the use of vector space duality and interior products becomes unmanageable.

The double algebra is the natural notation for computations with minors of a matrix; as the reader will recognize, such classical identities as Jacobi's and Sylvester's become trivial in the present contest (Section 5).

In the last section, the program of classical invariant theory, a program which remains largely unfulfilled, is outlined in the notation of Peano space. The murky versions of this problem given in the past are due to the lack of a suitable notation which would allow a clear conceptual formulation.

We have named these spaces after Peano, after becoming acquainted with Peano's remarkable book "*Calcolo geometrico secondo l'Ausdehnungslehre di H. Grassmann*," written in 1888 for Italian geometry students while the young author was still teaching at a military college. This book is, to our knowledge, the first that succeeds, at least in three dimensions, in axiomatizing join and meet by recognizing that a bracket must be given first.

The double algebra can in principle be put to use whenever exterior algebra methods are called for; the enrichment of structure that comes from using the meet may reveal aspects that would otherwise be difficult to single out.

2. PEANO SPACES

Like a Hilbert and a symplectic space, a Peano space is a vector space endowed with additional structure provided by a form with values in a field. Specifically, let K be an arbitrary field, whose values will be

sometimes called *scalars*, and let V be a vector space of dimension n over the field K , which will remain fixed throughout. We define a *bracket* (of step n) over the vector space V to be a non-degenerate alternating n -linear form defined over the vector space V , in symbols, a function

$$x_1, \dots, x_n \rightarrow [x_1, \dots, x_n] \in K$$

defined as the vectors $x_1, \dots, x_n$ range over the vector space V , with the following properties:

- (1) $[x_1, \dots, x_n] = 0$ if at least two among the x 's coincide;
- (2) for every $x, y \in V$ and $\alpha, \beta \in K$,

$$\begin{aligned} [x_1, \dots, x_{i-1}, \alpha x + \beta y, x_{i+1}, \dots, x_n] &= \alpha [x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n] \\ &\quad + \beta [x_1, \dots, x_{i-1}, y, x_{i+1}, \dots, x_n]; \end{aligned}$$

- (3) there exists a basis $\{b_1, \dots, b_n\}$ of V such that $[b_1, \dots, b_n] \neq 0$.

We define a *Peano space of step n* to be a pair $(V, [\])$, where V is a vector space of dimension n and $[\]$ is a bracket of step n over V . We shall denote a Peano space by the single letter V , leaving the bracket understood, whenever no confusion is possible. A non-degenerate multilinear alternating n -form on V is uniquely determined to within a non-zero multiplicative constant; however, the choice of this constant is crucial, as we shall see.

A Peano space can be viewed geometrically as a vector space in which an oriented volume element is specified. The bracket $[x_1, \dots, x_n]$ gives the volume of the parallelepiped whose sides are the x_i 's.

Thus one sees a parallel between the notion of Peano space and the notion of Hilbert space. In a Hilbert space, angles and length can be computed, and the geometry of Hilbert space is the geometry of angles and length. By contrast, in a Peano space only oriented volumes can be computed, and geometric notions definable in terms of oriented volumes are the theory of Peano spaces. We shall see in the sequel the remarkable wealth of geometric notions that can be so developed.

If V is a vector space of dimension n , a bracket on V of step n can be defined in several ways. The best known is simply to take a basis $e_1, \dots, e_n$ of V and, given vectors

$$x_i = \sum_j x_{ij} e_j, \quad i = 1, \dots, n,$$

to set

$$[x_1, \dots, x_n] = \det(x_{ij}).$$

A bracket can always be computed as a determinant. It is, however, against the spirit of the present work to view the bracket as a determinant in a specific basis, much as in the theory of Hilbert space it is distracting to view an inner product as being specifically evaluated in terms of finite or infinite sums, or integrals.

EXAMPLE (Wronskian space). Let

$$f^{(n)} + a_{n-1}(x)f^{(n-1)} + \cdots + a_0(x)f = 0$$

be a linear homogeneous differential equation of order n , where the functions $a_i(x)$ are of class $C^{(\infty)}[a, b]$. Let V be the vector space of all solutions of the differential equation. For fixed $c \in [a, b]$, define the bracket

$$\begin{aligned} [f_1, \dots, f_n]_c &= \text{Wr}(f_1, \dots, f_n)(c) \\ &= \det \begin{bmatrix} f_1(c) & \cdots & f_n(c) \\ f_1^{(1)}(c) & \cdots & f_n^{(1)}(c) \\ \vdots & \ddots & \vdots \\ f_1^{(n-1)}(c) & \cdots & f_n^{(n-1)}(c) \end{bmatrix} \end{aligned}$$

where $f_1, \dots, f_n \in V$. This defines a Peano space over V ; thus, the theory of Peano spaces is closely related to the classical theory of Wronskians. Oscillation properties, zeroes, total positivity and related questions (see, e.g., Karlin [10]) can be developed by the geometry of Peano spaces.

EXAMPLE (Subspaces and Quotients). Let W be a subspace of a Peano space V , and let $w_1, \dots, w_n$ be a basis of V such that $w_1, \dots, w_k$ is basis of W . We define the *restriction* of the Peano space to W to be the Peano space obtained by endowing W with the bracket

$$[x_1, \dots, x_k]_W = [x_1, \dots, x_k, w_{k+1}, \dots, w_n].$$

The bracket $[\]_W$ depends to within a multiplicative constant on a choice of the basis elements $w_{k+1}, \dots, w_n$.

Let W' be the subspace spanned by $w_{k+1}, \dots, w_n$. We define a Peano space on the quotient space V/W' by setting

$$[v_1, \dots, v_{n-k}/W'] = [x_1, \dots, x_k, w_{k+1}, \dots, w_n]$$

where v_i are vectors in V/W' and x_i is any vector that is mapped to v_i by the canonical map of V onto V/W' . Again, the bracket in the quotient Peano space depends on a choice of a basis of W' , but only to within a multiplicative constant.

3. THE JOIN

We develop the exterior algebra of a vector space in the context of Peano spaces. The facts are familiar, but it will be prudent to start from scratch.

Given a Peano space V of step n over the field K , let $S(V)$ be the free associative algebra, with unity, over K generated by the elements of V . For every integer k , $1 \leq k \leq n$, consider the subspace $N_k(V)$ of $S(V)$ of all vectors $f = \sum_i \alpha_i w_i$, with $w_i = x_1^{(i)} x_2^{(i)} \cdots x_k^{(i)}$, $x_j^{(i)} \in V$ and $\alpha_i \in K$, such that, for every $z_1, z_2, \dots, z_{n-k} \in V$,

$$\sum_i \alpha_i [x_1^{(i)}, x_2^{(i)}, \dots, x_k^{(i)}, z_1, z_2, \dots, z_{n-k}] = 0.$$

For $k > n$, let $N_k(V)$ be the subspace of $S(V)$ spanned by all words of length k , and let

$$N(V) = N_1(V) \oplus N_2(V) \oplus \cdots.$$

One verifies that $N(V)$ is an ideal of the algebra $S(V)$. The quotient algebra

$$G(V) = S(V)/N(V)$$

is the *exterior algebra of the Peano space V* .

Let $\phi: S(V) \rightarrow G(V)$ denote the canonical projection of $S(V)$ onto $G(V)$; if $x_1 x_2 \cdots x_k$ is a word in $S(V)$, $x_1, \dots, x_k \in V$, $k > 0$, we shall denote its image by

$$\phi(x_1 x_2 \cdots x_k) = x_1 \vee x_2 \vee \cdots \vee x_k$$

and, if $\phi(x_1 x_2 \cdots x_k) \neq 0$, we call it an *extensor of step k* .

For every k , $1 \leq k \leq n$, let $G_k(V)$ be the subspace of $G(V)$ generated by all extensors of step k ; in particular, $G_1(V) = V$. It is immediate that

$$G(V) = G_0(V) \oplus G_1(V) \oplus \cdots \oplus G_n(V)$$

where $G_0(V) = K$.

Any non-zero element of $G_0(V)$ is called an *extensor of step zero*.

The product in the exterior algebra is called the *join* and is denoted by the symbol $\vee$. We note that this usage is at variance with ordinary usage, where exterior multiplication is denoted by the wedge $\wedge$. This departure from common usage will be amply justified in the sequel.

We note that the join of two extensors is an extensor. Extensors will be denoted by capital letters. When an extensor A is written as

$$A = a_1 \vee a_2 \vee \cdots \vee a_k$$

we say that the linearly ordered set $\{a_1, a_2, \dots, a_k\}$ is a *representation* of the extensor A .

Not every element of $G(V)$ is an extensor; it is in fact a non-trivial problem to determine whether an element of $G(V)$ is an extensor, or more generally a linear combination of not more than i extensors. Extensors are also known as *decomposable antisymmetric tensors*, and a linear combination of extensors of step k is called an *antisymmetric tensor of step k* .

We collect some basic properties of the exterior algebra of a Peano space. Most of these properties are well known, and we shall omit the proofs.

3.1. PROPOSITION. (i) For every $i = 1, 2, \dots, k$, $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k$, $x, y \in V$, and for every $\alpha, \beta \in K$, we have:

$$\begin{aligned} & x_1 \vee \dots \vee x_{i-1} \vee (\alpha x + \beta y) \vee x_{i+1} \vee \dots \vee x_k \\ &= \alpha(x_1 \vee \dots \vee x_{i-1} \vee x \vee x_{i+1} \vee \dots \vee x_k) \\ &+ \beta(x_1 \vee \dots \vee x_{i-1} \vee y \vee x_{i+1} \vee \dots \vee x_k); \end{aligned}$$

(ii) for every permutation σ of $\{1, 2, \dots, k\}$,

$$x_{\sigma 1} \vee x_{\sigma 2} \vee \dots \vee x_{\sigma k} = \text{sign}(\sigma) x_1 \vee x_2 \vee \dots \vee x_k.$$

3.2. PROPOSITION. Let $A \in G_h(V)$, $B \in G_k(V)$; then

$$B \vee A = (-1)^{hk} A \vee B.$$

3.3. PROPOSITION. For every k , $0 \leq k \leq n$, the vector space $G_k(V)$ has dimension $\binom{n}{k}$. The exterior algebra $G(V)$ has dimension 2^n .

3.4. PROPOSITION. Let A be a subspace of V , of dimension $k > 0$; if $\{x_1, x_2, \dots, x_k\}$, $\{y_1, y_2, \dots, y_k\}$ are two bases of A , then

$$x_1 \vee x_2 \vee \dots \vee x_k = C y_1 \vee y_2 \vee \dots \vee y_k$$

for some non-zero scalar C .

Thus, every non-trivial subspace of V is uniquely represented (modulo a non-zero scalar) by a non-zero extensor, and vice versa, and the zero subspace is represented by the scalars. We say that such an extensor is *associated* to the subspace.

3.5. PROPOSITION. Let A, B be two subspaces of V , with associated extensors F and G , respectively. Then

- (i) $F \vee G = 0$ if and only if $A \cap B \neq \{0\}$;
 (ii) if $A \cap B = \{0\}$, then the extensor $F \vee G$ is associated to the subspace generated by $A \cup B$.

3.6. PROPOSITION. *Let V be a Peano space of step n over the field K , and let $G(V)$ be its exterior algebra. Choose a basis $\{a_1, a_2, \dots, a_n\}$ of V and let $S(a_1, a_2, \dots, a_n)$ be the free associative algebra with unity over K in the variables $a_1, a_2, \dots, a_n$. The algebra $G(V)$ is isomorphic to the quotient of the algebra $S(a_1, a_2, \dots, a_n)$ by the ideal generated by the following elements of $S(a_1, a_2, \dots, a_n)$:*

$$\begin{aligned} a_i^2, & \quad i = 1, 2, \dots, n; \\ a_i a_j + a_j a_i, & \quad i, j = 1, 2, \dots, n. \end{aligned}$$

The exterior algebra is well behaved relative to restrictions and quotients. We have in fact

3.7. PROPOSITION. *Let W be a subspace of the Peano space V , endowed with a restriction of the bracket, and let V/W be endowed with the quotient bracket. Then the exterior algebra of the Peano space W is naturally isomorphic to the restriction to W of the exterior algebra of the Peano space V , and the exterior algebra of the Peano space V/W is naturally isomorphic to the quotient of the exterior algebra of V by the ideal generated by the exterior algebra of W .*

4. THE MEET

We come now to the basic definition of the present work. We have seen that the operation " $\vee$ " in the exterior algebra of a Peano space (or of a vector space generally) represents algebraically the geometric operation of taking the join of two subspaces of a vector space, at least when such subspaces have the zero subspace as their intersection. It stands to reason that a similar operation should be defined for the intersection, or meet of two subspaces. This turns out to be true in the Peano space, where the bracket plays a crucial role in making such a definition possible. We shall fix henceforth a Peano space V of step n over a field K .

Given a representation of an extensor A

$$A = a_1 \vee \cdots \vee a_k$$

and an ordered r -tuple of non-negative integers $h_1, h_2, \dots, h_r$, such that $h_1 + h_2 + \cdots + h_r = k$, a *split* of class $(h_1, h_2, \dots, h_r)$ of the representation $A = a_1 \vee \cdots \vee a_k$ is an ordered r -tuple of extensors $(A_1, \dots, A_r)$ such that

- (i) $A_i = 1$ if $h_i = 0$ and $A_i = a_{i_1} \vee \cdots \vee a_{i_{h_i}}$, $i_1 < \cdots < i_{h_i}$ if $h_i \neq 0$;
- (ii) $A_i \vee A_j \neq 0$;
- (iii) $A_1 \vee A_2 \vee \cdots \vee A_r = \pm A$.

We shall denote by $\mathcal{S}(a_1, a_2, \dots, a_k; h_1, \dots, h_r)$ the (finite) set of all splits of class $(h_1, \dots, h_r)$ of the extensor A relative to the representation $A = a_1 \vee a_2 \vee \cdots \vee a_k$. We shall often use the symbol $\mathcal{S}(A; h_1, \dots, h_r)$ instead of $\mathcal{S}(a_1, a_2, \dots, a_k; h_1, \dots, h_r)$, when the choice of the representation of the extensor A is non-ambiguous in the context.

Given a split $A_1, A_2, \dots, A_r$ of an extensor A , we define

$$\begin{aligned} \text{sign}(A_1, A_2, \dots, A_r) &= 1 && \text{if } A_1 \vee A_2 \vee \cdots \vee A_r = A \\ &= -1 && \text{if } A_1 \vee A_2 \vee \cdots \vee A_r = -A. \end{aligned}$$

We shall now extend the bracket notation to include the case that the entries are extensors instead of vectors. This extension is not a mere formality, as we shall see. It leads to the discovery of some of the deeper identities of the double algebra of Peano spaces.

Let $A_1, A_2, \dots, A_h$ be extensors in a Peano space of step n . Choose representations

$$\begin{aligned} A_1 &= a_{11} \vee \cdots \vee a_{1s_1} \\ A_2 &= a_{21} \vee \cdots \vee a_{2s_2} \\ &\dots \\ A_h &= a_{h1} \vee \cdots \vee a_{hs_h} \end{aligned}$$

at will, and set

$$[A_1, A_2, \dots, A_h] = [a_{11}, \dots, a_{1s_1}, a_{21}, \dots, a_{2s_2}, \dots, a_{h1}, \dots, a_{hs_h}]$$

when $s_1 + s_2 + \cdots + s_h = n$, and

$$[A_1, A_2, \dots, A_h] = 0$$

otherwise.

This definition is verified to be independent of the choice of the representations of the extensors $A_1, A_2, \dots, A_h$.

4.1. PROPOSITION. *Let A and B be two extensors of steps $k, n-k$, respectively. Then*

$$[B, A] = (-1)^{k(n-k)} [A, B].$$

The proof is omitted.

We come now to a fundamental fact about Peano spaces:

4.2. PROPOSITION. *Let $a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_p$ be vectors of a Peano space V of step n , with $k + p > n$; set*

$$A = a_1 \vee a_2 \vee \dots \vee a_k, \quad B = b_1 \vee b_2 \vee \dots \vee b_p;$$

then, the following identity holds:

$$\begin{aligned} \sum_{(A_1, A_2) \in \mathcal{S}(A; n-p, k+p-n)} \text{sign}(A_1, A_2)[A_1, B] A_2 \\ = \sum_{(B_1, B_2) \in \mathcal{S}(B; k+p-n, n-k)} \text{sign}(B_1, B_2)[A, B_2] B_1. \end{aligned}$$

Proof. Consider the following functions:

$$f, g: V^{k+p} \rightarrow G_{k+p-n}(V)$$

defined as

$$\begin{aligned} f(a_1, \dots, a_k; b_1, \dots, b_p) &= \sum_{(A_1, A_2) \in \mathcal{S}(A; n-p, k+p-n)} \text{sign}(A_1, A_2)[A_1, B] A_2, \\ g(a_1, \dots, a_k; b_1, \dots, b_p) &= \sum_{(B_1, B_2) \in \mathcal{S}(B; k+p-n, n-k)} \text{sign}(B_1, B_2)[A, B_2] B_1, \end{aligned}$$

where $A = a_1 \vee \dots \vee a_k, B = b_1 \vee \dots \vee b_p$.

It is immediately seen that f and g are $(k+p)$ -multilinear functions; hence, they coincide if and only if they take the same values on any $(k+p)$ -tuple of vectors taken from a given basis $\{e_1, \dots, e_n\}$ of V . Furthermore, since f and g are alternating in the first k variables and in the last p variables, separately, it is sufficient to prove that

$$f(e_{i_1}, \dots, e_{i_k}; e_{j_1}, \dots, e_{j_p}) = g(e_{i_1}, \dots, e_{i_k}; e_{j_1}, \dots, e_{j_p})$$

in the case when $i_1 < \dots < i_k, j_1 < \dots < j_p$, and moreover, setting $d = k + p - n, i_1 = j_1, \dots, i_d = j_d$; straightforward computations show that, in this case,

$$f(e_{i_1}, \dots, e_{i_k}; e_{j_1}, \dots, e_{j_p}) = (-1)^{d(k-d)} [e_{i_{d+1}}, \dots, e_{i_k}, e_{j_1}, \dots, e_{j_p}] e_{i_1} \vee \dots \vee e_{i_d}$$

and

$$g(e_{i_1}, \dots, e_{i_k}; e_{j_1}, \dots, e_{j_p}) = [e_{i_1}, \dots, e_{i_k}, e_{j_{d+1}}, \dots, e_{j_p}] e_{i_1} \vee \dots \vee e_{i_d}. \quad \blacksquare$$

Given two extensors $A = a_1 \vee \dots \vee a_k, B = b_1 \vee \dots \vee b_p$, with $k, p \geq 1$, we define a binary operation, *meet*, written " $\wedge$," by setting:

(i) if $k + p < n$, $A \wedge B = 0$;

(ii) if $k + p \geq n$, then

$$A \wedge B = \sum_{(A_1, A_2)} \text{sign}(A_1, A_2)[A_1, B] A_2 = \sum_{(B_1, B_2)} \text{sign}(B_1, B_2)[A, B_2] B_1$$

where the summations range over the splits $\mathcal{S}(a_1, \dots, a_k; n - p, k + p - n)$ and $\mathcal{S}(b_1, \dots, b_p; k + p - n, n - k)$, respectively.

We remark that, if A has two representations, $a_1, \dots, a_k$ and $c_1, \dots, c_k$, then

$$\begin{aligned} & \sum_{(B_1, B_2)} \text{sign}(B_1, B_2)[a_1, \dots, a_k, B_2] B_1 \\ &= \sum_{(B_1, B_2)} \text{sign}(B_1, B_2)[c_1, \dots, c_k, B_2] B_1; \end{aligned}$$

this shows that the definition of $A \wedge B$ is independent of the representation of A . A similar argument shows that the meet is independent of the particular representation of B .

Let now A be an extensor of step k and B an extensor of step zero, that is, B equals a scalar $b \in K$. We define

(i) if $k < n$, $A \wedge B = 0$;

(ii) if $k = n$, $A \wedge B = b[A]$,

where $[A] = [a_1, \dots, a_n]$ in any representation $A = a_1 \vee \dots \vee a_n$ of the extensor A .

We have seen that the join of two extensors represents the geometric operation of joining two vector spaces. It stands to reason that the meet should play a similar role for the intersection of subspaces. This we now prove.

4.3. PROPOSITION. *Let A and B be associated to the subspaces X, Y of V , respectively. Then:*

If the union $X \cup Y$ spans the whole space V , and if $X \cap Y \neq 0$, then $A \wedge B$ is the extensor associated to the subspace $X \cap Y$.

Proof. Suppose that $X \cup Y$ spans V ; let k be the step of A , p the step of B , so that $k + p \geq n$. If $k + p \geq n$, the assertion holds trivially. Suppose $k + p > n$ and take a basis $\{c_1, c_2, \dots, c_d\}$ of the subspace $X \cap Y$ (here, $d = k + p - n$), and complete it first to a basis $\{c_1, \dots, c_d, a_1, \dots, a_{k-d}\}$ of X associated to A and then to a basis $\{c_1, \dots, c_d, b_1, \dots, b_{p-d}\}$ of Y associated to B , so that

$$\begin{aligned} A &= c_1 \vee \dots \vee c_d \vee a_1 \vee \dots \vee a_{k-d}, \\ B &= c_1 \vee \dots \vee c_d \vee b_1 \vee \dots \vee b_{p-d}. \end{aligned}$$

Now compute the meet $A \wedge B$ by splitting either A or B . The sum reduces to a single summand of the form

$$hc_1 \vee c_2 \vee \cdots \vee c_d$$

for some non-zero scalar h , and this extensor is associated to the intersection subspace $X \cap Y$. ■

We explicitly note that the preceding proposition implies that the meet of two extensors is again an extensor.

4.4. PROPOSITION. *Let A, B be two extensors of steps k and p , respectively. Then*

$$A \wedge B = (-1)^{(n-k)(n-p)} B \wedge A.$$

The proof follows from Proposition 3.1 and from the definition of meet. ■

4.5. PROPOSITION. *Let A, A_1, A_2, B be extensors such that $A = A_1 + A_2$; then*

$$A \wedge B = A_1 \wedge B + A_2 \wedge B.$$

Proof. If A is of step zero, the assertion is trivial; suppose $\text{step}(A) > 0$. Since $A = A_1 + A_2$ implies $[A, X] = [A_1, X] + [A_2, X]$ for every extensor X with $\text{step}(X) + \text{step}(A) = n$, we have, if $\text{step}(A) + \text{step}(B) > n$:

$$\begin{aligned} A \wedge B &= \sum_{(B_1, B_2)} \text{sign}(B_1, B_2) [A, B_2] B_1 \\ &= \sum_{(B_1, B_2)} \text{sign}(B_1, B_2) ([A_1, B_2] + [A_2, B_2]) B_1 \\ &= A_1 \wedge B + A_2 \wedge B. \end{aligned}$$

If $\text{step}(A) + \text{step}(B) = n$, the assertion holds immediately. ■

4.6. PROPOSITION. *Let A, B, C be extensors; then the associative law holds for the meet:*

$$(A \wedge B) \wedge C = A \wedge (B \wedge C).$$

Proof. Set $\text{step}(A) = i$, $\text{step}(B) = j$, $\text{step}(C) = k$; in order to avoid trivial cases, suppose $i, j, k > 0$ and $i + j > n$, $j + k > n$.

Computing the meets in succession by the definition, we have

$$\begin{aligned}
 (A \wedge B) \wedge C &= \sum_{(B_1, B_2) \in \mathcal{S}(B; i+j-n, n-i)} \text{sign}(B_1, B_2) [A, B_2] B_1 \wedge C \\
 &= \sum_{(B_1, B_2)} \sum_{(B_3, B_4) \in \mathcal{S}(B_1; n-k, i+j+k-2n)} \text{sign}(B_1, B_2) \\
 &\quad \times \text{sign}(B_3, B_4) [A, B_2] [B_3, C] B_4 \\
 &= \sum_{(B_3, B_4, B_2) \in \mathcal{S}(B; n-k, i+j+k-2n, n-i)} \text{sign}(B_3, B_4, B_2) [A, B_2] [B_3, C] B_4,
 \end{aligned}$$

whereas computing the right side we obtain:

$$\begin{aligned}
 A \wedge (B \wedge C) &= \sum_{(D_1, D_2) \in \mathcal{S}(B; n-k, j+k-n)} \text{sign}(D_1, D_2) [D_1, C] A \wedge D_2 \\
 &= \sum_{(D_1, D_2)} \sum_{(D_3, D_4) \in \mathcal{S}(D_2; i+j+k-2n, n-i)} \text{sign}(D_1, D_2) \\
 &\quad \times \text{sign}(D_3, D_4) [D_1, C] [A, D_4] D_3 \\
 &= \sum_{(D_1, D_3, D_4) \in \mathcal{S}(B; n-k, i+j+k-2n, n-i)} \text{sign}(D_1, D_3, D_4) [D_1, C] [A, D_4] D_3.
 \end{aligned}$$

Thus, the two sides are equal. ■

So far, we have defined the meet of extensors. We shall now extend the definition of meet to antisymmetric tensors generally. Thus, let

$$T = A + C$$

be a sum of extensors of step k , and let B be an extensor of step p . Then, recalling Proposition 4.5, the definition

$$T \wedge B = A \wedge B + C \wedge B$$

unequivocally defines the meet. By iterating this argument, one shows that the meet of any two elements of the exterior algebra $G(V)$ is well defined.

The vector space $G(V)$, endowed with the two operations of join and meet, will be called the *double algebra* of the Peano space V .

It is important to note that the meet of two extensors A and B can in general be computed in two ways, as indicated by either side of the identity in Proposition 4.2. The fact that there are two completely different ways of computing the meet of two extensors gives a great suppleness and power to

the double algebra of a Peano space. The computational power of the double algebra, first intuited by Hermann Grassmann, was obscured in the later development of exterior algebra in this century by the distinction between the exterior algebra of a vector space V and the exterior algebra of its dual space V^* . It has been our purpose to restore Grassmann's original ideas to their full geometrical power.

5. CAP-PRODUCTS

Let V be a Peano space of step n over the field K . We shall say that a linearly ordered basis $\{a_1, a_2, \dots, a_n\}$ of V is *unimodular* whenever

$$[a_1, a_2, \dots, a_n] = 1.$$

Let $\{a_1, a_2, \dots, a_n\}$ be any unimodular basis; the extensor

$$E = a_1 \vee a_2 \vee \cdots \vee a_n$$

will be called the *integral*. Clearly the integral is well defined, that is, it does not depend on the choice of a unimodular basis.

The integral behaves like an identity in the double algebra. Thus, for every extensor B with $\text{step}(B) > 0$, we have

$$B \vee E = 0, \quad B \wedge E = B,$$

while, for every scalar k , we have

$$k \vee E = kE, \quad k \wedge E = k.$$

For every n -tuple $(b_1, b_2, \dots, b_n)$ of vectors in V , we have the identity

$$b_1 \vee b_2 \vee \cdots \vee b_n = [b_1, b_2, \dots, b_n] E.$$

5.1. PROPOSITION. *Let A, B be extensors such that $\text{step}(A) + \text{step}(B) = n$. Then*

$$A \vee B = (A \wedge B) \vee E.$$

Proof. It follows immediately from the definition of meet and join that

$$A \wedge B = [A, B] \quad \text{and} \quad A \vee B = [A, B] E. \quad \blacksquare$$

Identities relating meets and joins are often far from obvious. The following is a sampling. It follows at once from the definitions:

5.2. PROPOSITION. *Let A, B, C be extensors such that $\text{step}(A) + \text{step}(B) + \text{step}(C) = n$; then*

$$A \wedge (B \vee C) = [A, B, C] = (A \vee B) \wedge C.$$

Similarly, we have

5.3. PROPOSITION. *Let A, B, C be extensors, with $\text{step}(A) = n$, $\text{step}(B) + \text{step}(C) = n$. Then*

$$A \wedge (B \vee C) = (A \wedge B \wedge C) \vee E.$$

Proof. As remarked above, $B \vee C = (B \wedge C) \vee E$. Hence,

$$A \wedge (B \vee C) = A \wedge ((B \wedge C) \vee E).$$

Since $B \wedge C$ is a scalar and since $\text{step}(A) = n$, the right side equals $(A \wedge B \wedge C) \vee E$. ■

An extensor of step $n - 1$ is called a *covector*. Covectors will be denoted by greek letters, say,

$$\alpha = a_1 \vee a_2 \vee \cdots \vee a_{n-1}.$$

All antisymmetric tensors of step $n - 1$ are decomposable, that is, they are covectors. More precisely:

5.4. PROPOSITION. *Every non-trivial linear combination of covectors is a covector.*

Proof. We can prove the assertion in the case of a linear combination

$$h\alpha + k\beta, \quad h, k \in K$$

of two covectors α and β , $\alpha \neq \beta$. In this case, the meet $\alpha \wedge \beta$ is an extensor of step $n - 2$. Clearly, there exist two vectors e, f such that

$$\alpha = (\alpha \wedge \beta) \vee e, \quad \beta = (\alpha \wedge \beta) \vee f;$$

hence,

$$h\alpha + k\beta = (\alpha \wedge \beta) \vee he + (\alpha \wedge \beta) \vee kf = (\alpha \wedge \beta) \vee (he + kf)$$

which is again a covector, as desired. ■

Similarly, one easily proves:

5.5. PROPOSITION. *Every extensor of step $k < n$ can be written as the meet of $n - k$ covectors.*

Our next topic is the reformulation in Peano spaces of some classical facts pertaining to the duality of vector spaces. In the present formulation, covectors replace linear functionals. Geometrically, this turns out to be an advantage, as we shall see.

We define the *cap-product* of an antisymmetric tensor E of step k with an antisymmetric tensor F of step $n - k$ by setting

$$(E, F)_k = E \wedge F.$$

Note that the right side is a scalar. Thus, the cap-product is a bilinear pairing of $G_k(V)$ and $G_{n-k}(V)$. We remark that

$$(F, E)_{n-k} = (-1)^{k(n-1)} (E, F)_k.$$

We shall now describe a remarkable pair of bases for the cap-product. To this end, we begin by developing a suitable notation for bases of the subspace $G_k(V)$. Given a linearly ordered basis $\{a_1, a_2, \dots, a_n\}$ of V and given a subset S of the linearly ordered set $\{1, 2, \dots, n\}$, say, $S = \{i_1, i_2, \dots, i_k\}$ with $i_1 < i_2 < \dots < i_k$, we write

$$a(S) = a_{i_1} \vee a_{i_2} \vee \dots \vee a_{i_k}.$$

For example, $a(\{i\}) = a_i$.

As S ranges over all k -element subsets of $\{1, 2, \dots, n\}$, the family of extensions $a(S)$ is a basis of the subspace $G_k(V)$.

The classical Laplace expansions for determinants can now be given an invariant formulation. Writing S^c for the complement of the set S in $\{1, 2, \dots, n\}$, we find:

5.6. PROPOSITION. *Let $\{a_1, a_2, \dots, a_n\}$ be a linearly ordered basis of V ; if S is a k -subset and T an $(n - k)$ -subset of $\{1, 2, \dots, n\}$, then*

$$\begin{aligned} (a(S), a(T))_k &= 0 & \text{if } T \neq S^c \\ &= [a(S), a(T)] & \text{if } T = S^c. \end{aligned}$$

Proof. Immediate from the definition of the meet. ■

Now set

$$\alpha(S) = [a(S), a(S^c)]^{-1} a(S^c) = (a(S), a(S^c))_k^{-1} a(S^c),$$

that is,

$$\alpha(S) = (-1)^{i_1 + \dots + i_k - k(k+1)/2} [a_1, \dots, a_n]^{-1} a_{p_1} \vee \dots \vee a_{p_{n-k}}$$

where $S = \{i_1, \dots, i_k\}$ and $S^c = \{p_1, \dots, p_{n-k}\}$. As S ranges over all k -subsets

of $\{1, 2, \dots, n\}$, we define the *associated cobasis* to the basis $a(S)$ of $G_k(V)$ to be the basis $\alpha(S)$ of $G_{n-k}(V)$. Note that the relation "associated cobasis" is not symmetric. We thus have:

5.7. PROPOSITION. *The associated cobasis $\alpha(S)$ of $G_{n-k}(V)$ is biorthogonal to the basis $a(S)$ of $G_k(V)$ relative to the cap-product, in symbols:*

$$\begin{aligned} (a(S), \alpha(T))_k &= 0 & \text{if } S \neq T \\ &= 1 & \text{if } S = T \end{aligned}$$

where $|S| = k$ and $|T| = k$.

5.8. COROLLARY. *For any antisymmetric tensor B of step k we have the expansion*

$$B = \sum_S (B, \alpha(S))_k a(S) = \sum_S (B \wedge \alpha(S)) a(S)$$

where S ranges over all k -subsets of $\{1, 2, \dots, n\}$.

In the special case $k = 1$, we define the *associated cobasis of covectors* of a given linearly ordered basis $\{a_1, a_2, \dots, a_n\}$ of V to be the set of covectors $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$, where

$$\alpha_i = [a_i, a_1, \dots, \hat{a}_i, \dots, a_n]^{-1} a_1 \vee \dots \vee \hat{a}_i \vee \dots \vee a_n,$$

so that $(a_i, \alpha_j)_1 = a_i \wedge \alpha_j = \delta_{ij}$.

We note in particular that, if $\{a_1, \dots, a_n\}$ is a unimodular basis, then its associated cobasis is $\{\alpha_1, \dots, \alpha_n\}$, with

$$\alpha_i = (-1)^{i-1} a_1 \vee \dots \vee \hat{a}_i \vee \dots \vee a_n.$$

5.9. THEOREM. *Let $\{a_1, a_2, \dots, a_n\}$ be a linearly ordered basis of V and $S = \{i_1, \dots, i_k\}$ a k -subset of $\{1, 2, \dots, n\}$, $i_1 < \dots < i_k$. Then*

$$\alpha(S) = \alpha_{i_1} \wedge \dots \wedge \alpha_{i_k}.$$

Proof. By induction on k . Set $k = 1$. Hence,

$$\begin{aligned} \alpha_i &= [a_i, a_1, \dots, \hat{a}_i, \dots, a_n]^{-1} a_1 \vee \dots \vee \hat{a}_i \vee \dots \vee a_n \\ &= (-1)^{i-1} [a_1, \dots, a_n]^{-1} a_1 \vee \dots \vee \hat{a}_i \vee \dots \vee a_n. \end{aligned}$$

Now, if $k > 1$, we have:

$$\begin{aligned}
 & (\alpha_{i_1} \wedge \cdots \wedge \alpha_{i_{k-1}}) \wedge \alpha_{i_k} \\
 &= (-1)^{(i_1-1) + \cdots + (i_{k-1}-k+1)} [a_1, \dots, a_n]^{-1} \\
 &\quad \times (a_1 \vee \cdots \vee \hat{a}_{i_1} \vee \cdots \vee \hat{a}_{i_{k-1}} \vee \cdots \vee a_n) \wedge ([a_{i_k}, a_1, \dots, \hat{a}_{i_k}, \dots, a_n]^{-1} \\
 &\quad \times a_1 \vee \cdots \vee \hat{a}_{i_k} \vee \cdots \vee a_n) \\
 &= (-1)^{(i_1-1) + \cdots + (i_{k-1}-k+1)} [a_1, \dots, a_n]^{-1} \\
 &\quad \times [a_{i_k}, a_1, \dots, \hat{a}_{i_k}, \dots, a_n]^{-1} [a_{i_k}, a_1, \dots, \hat{a}_{i_k}, \dots, a_n] (-1)^{i_k-k} \\
 &\quad \times a_1 \vee \cdots \vee \hat{a}_{i_1} \vee \cdots \vee \hat{a}_{i_k} \vee \cdots \vee a_n = \alpha(S). \quad \blacksquare
 \end{aligned}$$

5.10. COROLLARY. *Let $\{a_1, \dots, a_n\}$ be a basis of V . We have*

$$\alpha_1 \wedge \cdots \wedge \alpha_i = [a_1, \dots, a_n]^{-1} a_{i+1} \vee \cdots \vee a_n.$$

5.11. COROLLARY (Cauchy's adjugate Theorem). *Let $\{a_1, \dots, a_n\}$ be a basis of V ; then*

$$\alpha_1 \wedge \cdots \wedge \alpha_n = [a_1, \dots, a_n]^{-1}.$$

The following identity is the coordinate-free version of Jacobi's theorem on the adjugate.

5.12. PROPOSITION. *Let $\{a_1, \dots, a_n\}, \{b_1, \dots, b_n\}$ be two linearly ordered bases of V . Then, for k -tuples $i_1 < \cdots < i_k, j_1 < \cdots < j_k$, we have:*

$$\begin{aligned}
 & (a_{i_1} \vee \cdots \vee a_{i_k}) \wedge (\beta_{j_1} \wedge \cdots \wedge \beta_{j_k}) \\
 &= (-1)^{i_1 + \cdots + i_k + j_1 + \cdots + j_k} [a_1, \dots, a_n] [b_1, \dots, b_n]^{-1} \\
 &\quad \times (b_{q_1} \vee \cdots \vee b_{q_{n-k}}) \wedge (\alpha_{p_1} \wedge \cdots \wedge \alpha_{p_{n-k}})
 \end{aligned}$$

where $\{p_1, \dots, p_{n-k}\}, \{q_1, \dots, q_{n-k}\}$ are the complements in $\{1, 2, \dots, n\}$ of $\{i_1, \dots, i_k\}, \{j_1, \dots, j_k\}$, respectively, and $p_1 < \cdots < p_{n-k}, q_1 < \cdots < q_{n-k}$.

Proof. By Theorem 5.9, the statement can be rewritten as

$$\begin{aligned}
 \alpha(S) \wedge \beta(T) &= (-1)^{i_1 + \cdots + i_k + j_1 + \cdots + j_k} \\
 &\quad \times [a_1, \dots, a_n] [b_1, \dots, b_n]^{-1} b(T^c) \wedge \alpha(S^c),
 \end{aligned}$$

with S, T k -subsets of $\{1, 2, \dots, n\}$. $\blacksquare$

We shall now derive a classical identity of Sylvester, which will play a

crucial role in the sequel. Let $\phi: V \rightarrow V$ be an invertible linear transformation. It is easily shown that, setting

$$\tilde{\phi}(a_{i_1} \vee a_{i_2} \vee \cdots \vee a_{i_k}) = \phi(a_{i_1}) \vee \phi(a_{i_2}) \vee \cdots \vee \phi(a_{i_k})$$

for every k -subset of the basis $\{a_1, a_2, \dots, a_n\}$, we obtain an automorphism $\tilde{\phi}$ of the exterior algebra $G(V)$. (If the transformation ϕ is unimodular, that is, if $[\phi(a_1), \phi(a_2), \dots, \phi(a_n)] = 1$, then ϕ actually induces an automorphism of the double algebra, but this fact will not be used presently.) The restriction of $\tilde{\phi}$ to the subspace $G_k(V)$ will be written ϕ_k . It is the k th *induced transformation*. Writing $\det(\psi)$ for the determinant of a linear transformation ψ , we have:

5.13. THEOREM. *Let $\{a_1, \dots, a_n\}$ be a basis of V , $[a_1, \dots, a_n] = 1$ and let $\phi: V \rightarrow V$ be an invertible linear transformation. Then, for every integer k between 0 and n , the following identity holds:*

$$\det(\phi_k) \det(\phi_{n-k}) = [\phi(a_1), \dots, \phi(a_n)] \binom{n}{k}.$$

Proof. For every S, T subsets of $\{1, 2, \dots, n\}$ with $|S| = k, |T| = k$, we have:

$$(\phi_k(a(S)), \phi_{n-k}(\alpha(T)))_k = [\phi(a_1), \dots, \phi(a_n)](a(S), \alpha(T))_k;$$

set now

$$\psi = [\phi(a_1), \dots, \phi(a_n)]^{-1} \phi_{n-k};$$

then, we get

$$\begin{aligned} (\phi_k(a(S)), \psi(\alpha(T)))_k &= 1 && \text{if } T = S \\ &= 0 && \text{otherwise.} \end{aligned}$$

This implies that the transformation ψ is the inverse of the adjoint of ϕ_k , and hence

$$\det(\phi_k) \det(\psi) = 1;$$

since

$$\det(\phi_{n-k}) = [\phi(a_1), \dots, \phi(a_n)] \binom{n}{k} \det(\psi),$$

the statement follows. ■

By way of application of the preceding identities, we now derive the matrix version of Jacobi's theorem on the adjugate and Sylvester's theorem.

First, we note that the bracket can be expressed as a determinant whenever a basis $\eta_1, \dots, \eta_n$ of covectors is given such that $\eta_1 \wedge \dots \wedge \eta_n = 1$; more precisely, for every n -tuple $a_1, \dots, a_n$ in V , we have

$$[a_1, \dots, a_n] = \det([a_i, \eta_j])_{i,j=1, \dots, n}. \quad (*)$$

Indeed, if $\{e_1, \dots, e_n\}$ is the basis of V such that $(e_i, \eta_j)_1 = \delta_{i,j}$ for every $i, j = 1, 2, \dots, n$, by Corollary 5.11 we obtain

$$[e_1, \dots, e_n] = 1 = \det([e_i, \eta_j])_{i,j=1, \dots, n};$$

now, $(*)$ holds by multilinearity.

Conversely, if $A = (a_{ij})$ is an $n \times n$ matrix, setting

$$a_i = \sum_j a_{ij} e_j$$

we obtain an expression of the determinant of A as a bracket, namely,

$$\det(A) = [a_1, a_2, \dots, a_n].$$

On the other hand, if $e_1, \dots, e_n$ is a unimodular basis of V , then, for every n -tuple of covectors $\alpha_1, \dots, \alpha_n$, we have

$$\alpha_1 \wedge \dots \wedge \alpha_n = \det([e_i, \alpha_j])_{i,j=1, \dots, n}.$$

By a similar argument, we get the identity:

5.14. PROPOSITION. *Let $\eta_1, \dots, \eta_n$ be covectors such that $\eta_1 \wedge \dots \wedge \eta_n = 1$; for every $k \leq n$, choose k vectors $a_1, \dots, a_k$ in V ; then*

$$[a_1, \dots, a_k, \eta_{j_1}, \dots, \eta_{j_k}] = \det([a_i, \eta_j])_{i=1, \dots, k; j=j_1, \dots, j_k}.$$

The preceding remarks allow us to obtain matrix translation of identities in the double algebra. We give some simple examples.

Binet's theorem states that, if M, N are $n \times n$ matrices, then

$$\det(M \times N) = \det M \cdot \det N.$$

Now, if $M = (m_{ij}), N = (n_{ij})$, then, setting

$$a_i = \sum_{k=1}^n m_{ik} e_k, \quad \beta_i = \sum_{k=1}^n n_{kj} \eta_k, \quad i, j = 1, \dots, n,$$

where $\{e_1, \dots, e_n\}$ is a unimodular basis of V and $\{\eta_1, \dots, \eta_n\}$ its associated cobasis, we find:

$$(a_1 \vee \dots \vee a_n) \wedge (\beta_1 \wedge \dots \wedge \beta_n) = \det([a_i, \beta_j]) = \det(M \times N)$$

and on the other hand,

$$\begin{aligned} (a_1 \vee \cdots \vee a_n) \wedge (\beta_1 \wedge \cdots \wedge \beta_n) &= [a_1, \dots, a_n](\beta_1 \wedge \cdots \wedge \beta_n) \\ &= \det M \cdot \det N, \end{aligned}$$

as desired.

Recall that, given an $n \times n$ matrix $A = (a_{ij})$, for every $k \times k$ submatrix $M = (a_{ij})$, $i = i_1, \dots, i_k$, $j = j_1, \dots, j_k$, the *complementary minor* of M is meant as the scalar

$$(-1)^{i_1 + \cdots + i_k + j_1 + \cdots + j_k} \det(a_{pq}), \quad p = p_1, \dots, p_{n-k}, q = q_1, \dots, q_{n-k},$$

where $\{p_1, \dots, p_{n-k}\}$, $\{q_1, \dots, q_{n-k}\}$ are the complements in $\{1, 2, \dots, n\}$ of $\{i_1, \dots, i_k\}$, $\{j_1, \dots, j_k\}$, respectively.

If $A = (a_{ij})$ is a non-singular matrix, let $A^{-1} = (\tilde{a}_{ij})$ be its inverse; then, the (i, j) th entry of A^{-1} is the complementary minor of a_{ij} multiplied by $(\det A^{-1})$. To see this, set

$$a_i = \sum_j a_{ij} e_j, \quad i = 1, \dots, n,$$

where $\{e_1, \dots, e_n\}$ is a unimodular basis of V , and let $\{\eta_1, \dots, \eta_n\}$, $\{\alpha_1, \dots, \alpha_n\}$ be the associated cobases of $\{e_1, \dots, e_n\}$, $\{a_1, \dots, a_n\}$, respectively. Since

$$(e_i, \eta_j)_1 = \delta_{i,j} = (a_i, \alpha_j)_1,$$

the matrix $B = (b_{ij}) = ((e_j, \alpha_i)_1)$ satisfies the identity $B = (A^{-1})^T$. Then

$$A^{-1} = (\tilde{a}_{ij}) = (e_i \wedge \alpha_j)$$

and, by Proposition 5.12,

$$\begin{aligned} e_i \wedge \alpha_j &= (-1)^{i+j} [a_1, \dots, a_n] \\ &\quad \times (a_1 \vee \cdots \vee \hat{a}_j \vee \cdots \vee a_n) \wedge \eta_1 \wedge \cdots \wedge \hat{\eta}_i \wedge \cdots \wedge \eta_n, \end{aligned}$$

which verifies the assertion.

The *adjugate matrix* of a non-singular matrix A is the matrix $B = (b_{ij})$ such that b_{ij} is the complementary minor of a_{ij} . Hence, $b_{ij} = \det(A) \cdot \tilde{a}_{ji}$, where $A^{-1} = (\tilde{a}_{ij})$. Jacobi's identity asserts that, for every $k \times k$ submatrix $M = (a_{ij})$, $i = i_1, \dots, i_k$, $j = j_1, \dots, j_k$ of A , the complementary minor of M equals

$$\det(A)^{-k+1} \det(b_{ij}), \quad i = i_1, \dots, i_k, j = j_1, \dots, j_k.$$

To see this, let $S = \{i_1, \dots, i_k\}$, $T = \{j_1, \dots, j_k\}$ and $S^c = \{p_1, \dots, p_{n-k}\}$, $T^c =$

$\{q_1, \dots, q_{n-k}\}$, with $p_1 < \dots < p_{n-k}, q_1 < \dots < q_{n-k}$; then, by Proposition 5.12, we have

$$\begin{aligned} & (-1)^{i_1 + \dots + i_k + j_1 + \dots + j_k} (a_{p_1} \vee \dots \vee a_{p_{n-k}}) \wedge \eta_{q_1} \wedge \dots \wedge \eta_{q_{n-k}} \\ &= [a_1, \dots, a_n] (e_{j_1} \vee \dots \vee e_{j_k}) \wedge \alpha_{i_1} \wedge \dots \wedge \alpha_{i_k} \\ &= [a_1, \dots, a_n] \det(e_j \wedge \alpha_i)_{j=1, \dots, j_k, i=1, \dots, i_k} \\ &= [a_1, \dots, a_n] \det(\tilde{a}_{ji})_{j=1, \dots, j_k, i=1, \dots, i_k} \\ &= [a_i, \dots, a_n]^{1-k} \det(b_{ij})_{i=1, \dots, i_k, j=1, \dots, j_k}, \end{aligned}$$

an easy computation indeed.

Lastly, we derive the celebrated theorem of Sylvester on compound determinants, to wit: fix an integer $k, 1 \leq k \leq n$, and let $I_1, I_2, \dots, I_{\binom{n}{k}}$ be the $\binom{n}{k}$ -tuple of increasing k -tuples of $\{1, 2, \dots, n\}$, lexicographically ordered. Set

$$A_{I_p, I_q} = \det(a_{ij})_{i \in I_p, j \in I_q}, \quad p, q = 1, 2, \dots, \binom{n}{k}.$$

Then

$$\det(A_{I_p, I_q})_{p, q = 1, 2, \dots, \binom{n}{k}} = (\det(A))^{\binom{n-1}{k-1}}.$$

The proof is practically immediate. Let $\{a_1, \dots, a_n\}$ be a unimodular basis of V and $\phi: V \rightarrow V$ be the linear transformation with matrix A with respect to $\{a_1, \dots, a_n\}$. The matrix

$$M = (A_{I_p, I_q}), \quad p, q = 1, 2, \dots, \binom{n}{k},$$

is the matrix of the k th induced transformation of ϕ relative to the linearly ordered basis $\{a(I_1), \dots, a(I_{\binom{n}{k}})\}$ of $G_k(V)$. Recall that $\det(A)$ is an irreducible polynomial in the a_{ij} 's; since $\det(M)$ is a polynomial of degree $\binom{n-1}{k-1}$ in the a_{ij} 's, Theorem 5.13 gives the assertion at once.

6. THE HODGE STAR OPERATOR

It is by now obvious that the meet operation defines a second exterior algebra structure on the vector space $G(V)$. Our present objective is to relate the two exterior algebras, defined by join and meet, by a suitable duality operator, which turns out to be the Hodge star operator (see, e.g., [12]). In the context of Peano spaces the theory of this operator acquires a crystalline simplicity, and the algebraic system obtained by meet, join and star operator is at last seen to be the much-sought-after linear analog of the Boolean algebra of sets with intersection, union and complement, an analogy which was first intuited by Grassmann and later pursued, with partial success, by Clifford, Peano, Burali-Forti and A. N. Whitehead.

Let $\{a_1, \dots, a_n\}$ be a linearly ordered basis of V ; we define the *Hodge star operator relative to the basis* $\{a_1, \dots, a_n\}$ to be the (unique) linear operator

$$*: G(V) \rightarrow G(V)$$

such that, for every subset S of $\{1, 2, \dots, n\}$,

$$* 1 = E$$

$$* a(S) = \alpha(S),$$

that is, if $S = \{i_1, \dots, i_k\}$, with $i_1 < \dots < i_k$,

$$* a_{i_1} \vee \dots \vee a_{i_k} = (-1)^{i_1 + \dots + i_k - k(k+1)/2} [a_1, \dots, a_n]^{-1} a_{p_1} \vee \dots \vee a_{p_{n-k}}$$

where $\{p_1, \dots, p_{n-k}\} = S^c$ and $p_1 < \dots < p_{n-k}$. By Theorem 5.9, this definition is equivalent to setting

$$* 1 = E$$

$$* a_{i_1} \vee \dots \vee a_{i_k} = \alpha_{i_1} \wedge \dots \wedge \alpha_{i_k},$$

where $\{\alpha_1, \dots, \alpha_n\}$ is the associated cobasis of covectors of $\{a_1, \dots, a_n\}$. We are therefore led to the following

6.1. PROPOSITION. *A Hodge star operator is an algebra isomorphism between the exterior algebra of the join $(G(V), \vee)$ and the exterior algebra of the meet $(G(V), \wedge)$. Moreover, it maps the set of extensors of $G_k(V)$ onto the set of extensors of $G_{n-k}(V)$.*

Proof. The first assertion is obvious by preceding remarks. To prove the second assertion, it suffices to recall that a Hodge star operator maps vectors to covectors and, being an algebra isomorphism, it maps joins of vectors to meets of covectors. ■

We shall use double algebra techniques to relate the two Hodge star operators defined relative to two different bases. This is a generalization of a result of Hodge:

6.2. THEOREM. *Let $\{a_1, \dots, a_n\}$ be a linearly ordered basis of V , with associated cobasis $\{\alpha_1, \dots, \alpha_n\}$. Consider another linearly ordered basis $\{b_1, \dots, b_n\}$ of V ; the Hodge star operator relative to $\{b_1, \dots, b_n\}$ coincides with the Hodge star operator relative to $\{a_1, \dots, a_n\}$ if and only if the matrix $B = ([b_i, \alpha_j])$ is orthogonal.*

Proof. Recall that, for every i :

$$b_i = \sum_j [b_i, \alpha_j] a_j.$$

Let $\{\beta_1, \dots, \beta_n\}$ be the associated cobasis of $\{b_1, \dots, b_n\}$ and set $M = (m_{ij})$, with

$$\beta_i = \sum_j m_{ij} \alpha_j.$$

Since $(a_i, \alpha_j)_1 = \delta_{ij} = (b_i, \beta_j)_1$, it follows $M = (B^T)^{-1}$.

Since the Hodge star operators relative to $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_n\}$ coincide if and only if $B = M$, the assertion follows. ■

More can be said about the Hodge star operator when the basis $\{a_1, \dots, a_n\}$ is unimodular. In this case, the Hodge star operator implements the duality between meet and join. This duality turns out to satisfy a series of identities reminiscent of set-theoretic duality for Boolean operations:

6.3. THEOREM. *Let the linearly ordered basis $\{a_1, \dots, a_n\}$ be unimodular. Then, the Hodge star operator $*$ relative to $\{a_1, \dots, a_n\}$ satisfies the following:*

- (i) *$*$ maps extensors of step k to extensors of step $n - k$;*
- (ii) *$*(x \vee y) = (*x) \wedge (*y)$ and $*(x \wedge y) = (*x) \vee (*y)$, for every $x, y \in G(V)$;*
- (iii) *$*(1) = E$ and $*(E) = 1$;*
- (iv) *$__(*x) = (-1)^{k(n-k)} x$, for every $x \in G_k(V)$.*

Proof. When $\{a_1, \dots, a_n\}$ is a unimodular basis, Corollary 5.11 gives at once

$$*(E) = *(a_1 \vee \dots \vee a_n) = \alpha_1 \wedge \dots \wedge \alpha_n = 1.$$

Given two extensors A, B of steps k and $n - k$, respectively, we have

$$(A, B)_k E = A \vee B$$

and, applying $*$ to both sides,

$$\begin{aligned} (A, B)_k &= (*A) \wedge (*B) = (*A, *B)_{n-k} \\ &= (-1)^{k(n-1)} (*B, *A)_k. \end{aligned}$$

In particular, for every S, T k -subsets of $\{1, 2, \dots, n\}$, we get

$$\begin{aligned} (-1)^{k(n-1)} (*\alpha(S), *a(T))_k &= (a(T), \alpha(S))_k = 1 && \text{if } S = T \\ &= 0 && \text{otherwise} \end{aligned}$$

which implies

$$*\alpha(S) = (-1)^{k(n-1)} a(S). \quad \blacksquare$$

By use of a Hodge star operator relative to a unimodular basis, we can state the *duality principle*: for any identity $p(A_1, \dots, A_p) = 0$ between extensors A_i of steps k_i , joins and meets holding in the double algebra, the identity $\hat{p}(\hat{A}_1, \dots, \hat{A}_p) = 0$ obtained by exchanging joins with meets and replacing A_i by an extensor $\hat{A}_i$ of step $n - k_i$ also holds.

7. ALTERNATIVE LAWS

We come now to the heart of the present work, that is, to the statement and proof of the deeper identities relating meet and join. We propose the term *alternative laws* for these identities. The coordinate-free and duality-free notation we have introduced, which could heretofore be seen as a mere elegant variant of exterior algebra, at this point becomes an indispensable device in the very statement of the alternative laws. In fact, the notational awkwardness of previous notations, notably the stubborn insistence in preserving vector space duality, cost what may, and the concomitant difficulties in the handling of inner products, all but prevented a clear statement, let alone an understanding of the full strength of the alternative laws.

Roughly, an alternative law is an identity which can be used in simplifying expressions containing joins and meets of extensors of different step.

7.1. PROPOSITION. *Let $a_1, a_2, \dots, a_k$ be vectors and let $\gamma_1, \gamma_2, \dots, \gamma_s$ be covectors. Set $A = a_1 \vee a_2 \vee \dots \vee a_k$. If $k \geq s$, then:*

$$\begin{aligned} A \wedge (\gamma_1 \wedge \dots \wedge \gamma_s) \\ &= \sum_{(A_1, \dots, A_{s+1}) \in \mathcal{S}(A; 1, \dots, 1, k-s)} \text{sign}(A_1, \dots, A_{s+1}) \\ &\quad \times [A_1, \gamma_1][A_2, \gamma_2] \dots [A_s, \gamma_s] A_{s+1}. \end{aligned}$$

Proof. The proof relies essentially on the associativity of the meet operation. Indeed, associating γ_1 to A and computing the meet by splitting the extensor A , we obtain

$$\begin{aligned} A \wedge (\gamma_1 \wedge \dots \wedge \gamma_s) \\ &= (A \wedge \gamma_1) \wedge (\gamma_2 \wedge \dots \wedge \gamma_s) \\ &= \left(\sum_{(A_1, B_2) \in \mathcal{S}(A; 1, k-1)} \text{sign}(A_1, B_2) [A_1, \gamma_1] B_2 \right) \wedge (\gamma_2 \wedge \dots \wedge \gamma_s). \end{aligned}$$

Again associating B_2 with γ_2 and computing the meet $B_2 \wedge \gamma_2$ by splitting the extensor B_2 we see that the above expression equals

$$\begin{aligned} & \sum_{(A_1, B_2) \in \mathcal{S}(A; 1, k-1)} \text{sign}(A_1, B_2) [A_1, \gamma_1] \\ & \quad \times \sum_{(A_2, B_3) \in \mathcal{S}(B_2; 1, k-2)} \text{sign}(A_2, B_3) [A_2, \gamma_2] B_3 \wedge (\gamma_3 \wedge \cdots \wedge \gamma_s) \\ & = \sum_{(A_1, A_2, B_3) \in \mathcal{S}(A; 1, 1, k-2)} \text{sign}(A_1, A_2, B_3) [A_1, \gamma_1] \\ & \quad \times [A_2, \gamma_2] B_3 \wedge (\gamma_3 \wedge \cdots \wedge \gamma_s). \end{aligned}$$

Continuing in this way, we obtain the expression on the right side of the asserted identity. ■

EXAMPLE. $(a_1 \vee a_2 \vee a_3) \wedge (\gamma_1 \wedge \gamma_2) = [a_1, \gamma_1][a_2, \gamma_2] a_3 - [a_1, \gamma_1][a_3, \gamma_2] a_2 - [a_2, \gamma_1][a_1, \gamma_2] a_3 + [a_2, \gamma_1][a_3, \gamma_2] a_1 + [a_3, \gamma_1][a_1, \gamma_2] a_2 - [a_3, \gamma_1][a_2, \gamma_2] a_1.$

7.2. COROLLARY. Let $a_1, \dots, a_n$ be vectors, $\beta_1, \dots, \beta_n$ covectors; then,

$$(a_1 \vee \cdots \vee a_n) \wedge (\beta_1 \wedge \cdots \wedge \beta_n) = \det[a_i, \beta_j]_{i,j=1,2,\dots,n}.$$

In the sequel, it will be useful to have the notion of split of an extensor written as meet of covectors; namely, if A is an extensor and

$$A = \alpha_1 \wedge \cdots \wedge \alpha_k$$

with $\alpha_1, \dots, \alpha_k$ covectors, given an ordered s -tuple of non-negative integers $k_1, \dots, k_s$ such that $k_1 + \cdots + k_s = k$, a *cosplit* of class $(k_1, \dots, k_s)$ of the representation $A = \alpha_1 \wedge \cdots \wedge \alpha_k$ is an ordered s -tuple of extensors such that

- (i) $A_i = E$ if $k_i = 0$ and $A_i = \alpha_{i_1} \wedge \cdots \wedge \alpha_{i_{k_i}}$, $i_1 < \cdots < i_{k_i}$ if $k_i \neq 0$;
- (ii) $A_i \wedge A_j \neq 0$;
- (iii) $A_1 \wedge \cdots \wedge A_s = \pm A$.

We denote by $\mathcal{C}(\alpha_1 \wedge \cdots \wedge \alpha_k; k_1, \dots, k_s)$ the set of all cosplits of class $(k_1, \dots, k_s)$ of the ordered set of covectors $\{\alpha_1, \dots, \alpha_k\}$. We shall use the notation $\mathcal{C}(A; k_1, \dots, k_s)$ in place of $\mathcal{C}(\alpha_1 \wedge \cdots \wedge \alpha_k; k_1, \dots, k_s)$, when no confusion is possible. Given an ordered s -tuple $(A_1, \dots, A_s)$ of extensors belonging to $\mathcal{C}(A; k_1, \dots, k_s)$, we define, much as before,

$$\begin{aligned} \text{sign}(A_1, \dots, A_s) &= 1 & \text{if } A_1 \wedge \cdots \wedge A_s = A \\ &= -1 & \text{if } A_1 \wedge \cdots \wedge A_s = -A. \end{aligned}$$

In this notation, the dual of the preceding Proposition is easily stated and even easier proved:

7.3. PROPOSITION. *Let $\alpha_1, \dots, \alpha_k$ be covectors and let $b_1, \dots, b_s$ be vectors, with $k > s$. Set $A = \alpha_1 \wedge \dots \wedge \alpha_k$. Then*

$$\begin{aligned} A \vee (b_1 \vee \dots \vee b_s) \\ = \sum_{(A_1, \dots, A_{s+1}) \in \mathcal{C}(A; 1, \dots, 1, k-s)} \text{sign}(A_1, \dots, A_{s+1}) \\ \times [A_1, b_1][A_2, b_2] \dots [A_s, b_s] A_{s+1}. \end{aligned}$$

Proof. Simply apply the Hodge star operator relative to a unimodular basis to both sides of the identity in Proposition 7.1. ■

There is no reason why the associative law for the meet should be applied stepwise to one extensor at a time, as in Proposition 7.1. Any bracketing scheme for extensors leads to a corresponding identity. One obtains in this way remarkable variations on the theme of Laplace expansion. Here is a sampling:

7.4. THEOREM. *Let $a_1, \dots, a_k$ be vectors and $\gamma_1, \dots, \gamma_s$ covectors. Set $A = a_1 \vee \dots \vee a_k$. If $k \geq s$, given a decomposition $s = i_1 + \dots + i_p$ of the integer s into a linearly ordered sequence $(i_1, \dots, i_p)$ of positive integers, we have:*

$$\begin{aligned} A \wedge (\gamma_1 \wedge \dots \wedge \gamma_s) \\ = \sum_{(A_1, \dots, A_{p+1}) \in \mathcal{S}(A; i_1, \dots, i_p, k-s)} \text{sign}(A_1, \dots, A_{p+1}) \\ \times [A_1, \gamma_1 \wedge \dots \wedge \gamma_{i_1}] \dots [A_p, \gamma_{s-i_p+1} \wedge \dots \wedge \gamma_s] A_{p+1}. \end{aligned}$$

Proof. Simply apply the associative law for the meet, that is:

$$\begin{aligned} A \wedge (\gamma_1 \wedge \dots \wedge \gamma_s) \\ = (A \wedge (\gamma_1 \wedge \dots \wedge \gamma_{i_1})) \wedge ((\gamma_{i_1} \wedge \dots \wedge \gamma_{i_1+i_2}) \\ \wedge \dots \wedge (\gamma_{s-i_p+1} \wedge \dots \wedge \gamma_s)). \end{aligned}$$

Now, compute the meet by successively splitting the extensor A . ■

Dualizing by a Hodge star operator relative to a unimodular basis, one obviously obtains:

7.5. PROPOSITION. *Let $\alpha_1, \dots, \alpha_k$ be covectors and $b_1, \dots, b_s$ vectors. Set $A = \alpha_1 \wedge \dots \wedge \alpha_k$. If $k > s$, for every decomposition $s = i_1 + \dots + i_p$, we have:*

$$\begin{aligned} & A \vee (b_1 \vee \dots \vee b_s) \\ &= \sum_{(A_1, \dots, A_{p+1}) \in \mathcal{C}(A; i_1, \dots, i_p, k-s)} \text{sign}(A_1, \dots, A_{p+1}) \\ & \quad \times [A_1, b_1 \vee \dots \vee b_{i_1}] \dots [A_p, b_{s-i_p+1} \vee \dots \vee b_s] A_{p+1}. \end{aligned}$$

The next identity lies slightly deeper. It can be viewed as the result of partitioning a sequence of covectors and bracketing the covectors within each block. Note that for the first time entire extensors are shuffled in the computation of a meet, with no need to split each extensor into a meet of covectors or a join of vectors.

7.6. PROPOSITION. *Let $A, B_1, \dots, B_p$ be extensors, such that $\text{step}(B_i) = n - s_i < n$ and $k = \text{step}(A) \geq s_1 + \dots + s_p$. Then*

$$\begin{aligned} & A \wedge (B_1 \wedge \dots \wedge B_p) \\ &= \sum_{(A_1, \dots, A_{p+1}) \in \mathcal{S}(A; s_1, \dots, s_p, k - \sum_i s_i)} \text{sign}(A_1, \dots, A_{p+1}) \\ & \quad \times [A_1, B_1] \dots [A_p, B_p] A_{p+1}. \end{aligned}$$

Proof. By the associativity of the meet, we have

$$A \wedge (B_1 \wedge \dots \wedge B_p) = (A \wedge B_1) \wedge (B_2 \wedge \dots \wedge B_p).$$

Now, compute the first meet on the right by splitting the extensor A . It equals

$$\sum_{(A_1, C_2) \in \mathcal{S}(A; s_1, k - s_1)} \text{sign}(A_1, C_2) [A_1, B_1] C_2.$$

Again associating C_2 with B_2 and computing the meet by splitting C_2 , we find

$$\begin{aligned} & A \wedge (B_1 \wedge \dots \wedge B_p) \\ &= \sum_{(A_1, A_2, C_3) \in \mathcal{S}(A; s_1, s_2, k - s_1 - s_2)} \text{sign}(A_1, A_2, C_3) \\ & \quad \times [A_1, B_1] [A_2, B_2] (C_3 \wedge (B_3 \wedge \dots \wedge B_p)). \end{aligned}$$

Iterating this argument, the assertion follows. ■

The dual identity is again obtained from the preceding by duality:

7.7. PROPOSITION. *Let $A_1, B_1, \dots, B_p$ be extensors such that $\text{step}(B_i) = s_i > 0$ and $k = \text{step}(A) < n - s_1 - \dots - s_p$. Then*

$$\begin{aligned} A \vee (B_1 \vee \dots \vee B_p) \\ = \sum_{(A_1, \dots, A_{p+1}) \in \mathcal{C}(A; s_1, \dots, s_p, n - k - \sum s_i)} \text{sign}(A_1, \dots, A_{p+1}) \\ \times [A_1, B_1] \cdots [A_p, B_p] A_{p+1}. \end{aligned}$$

The following identity owes its simplicity to the naturalness of the present notation. If nothing else, it should win over the reader to the notion of Peano space. The reader may find it instructive to rewrite it and to attempt a proof in matrix notation.

7.8. COROLLARY. *Let A, B, C be extensors, with $\text{step}(A) = n$, $\text{step}(B) = k > 0$ and $\text{step}(C) = n - k > 0$. Then*

$$[A][B, C] = \sum_{(A_1, A_2) \in \mathcal{S}(A; n - k, k)} \text{sign}(A_1, A_2)[A_1, B][A_2, C].$$

Proof. Write $[A][B, C] = A \wedge (B \wedge C)$ and apply Proposition 4.2. ■

EXAMPLE (Pappus theorem). Let a, b, c, a', b', c' be vectors in a Peano space of step 3. We prove the identity

$$\begin{aligned} & [((b \vee c') \wedge (b' \vee c)), ((c \vee a') \wedge (c' \vee a)), ((a \vee b') \wedge (a' \vee b))] \\ & = [b, c', b'][c', a, a'][c, b', a'][a, c, b] \\ & \quad - [b', c, b][c, a', a][c', b, a][a', c', b']. \end{aligned}$$

Indeed, the left side can be rewritten as

$$(((b \vee c') \wedge (b' \vee c)) \vee ((c \vee a') \wedge (c' \vee a))) \wedge ((a \vee b') \wedge (a' \vee b));$$

computing the outermost meet by splitting the extensor on the left, we get

$$\begin{aligned} & (((b \vee c') \wedge (b' \vee c)) \wedge (a \vee b')) \vee (((c \vee a') \wedge (c' \vee a)) \wedge (a' \vee b)) \\ & \quad - (((b \vee c') \wedge (b' \vee c)) \wedge (a' \vee b)) \vee (((c \vee a') \wedge (c' \vee a)) \wedge (a \vee b')); \end{aligned}$$

expanding the outermost meet by Proposition 7.5, this equals

$$\begin{aligned} & [b', c, a][b, c', b'][c', a, a'][c, a', b] \\ & \quad - [b, c', a'][b', c, b][c, a', a][c', a, b']; \end{aligned}$$

now, by Corollary 7.8, we expand

$$\begin{aligned} [b', c, a][c, a', b] &= [a, c, b'][a', c, b] \\ &= [a, c, a'][b', c, b] + [c, b', a'][a, c, b], \\ [b, c', a'][c', a, b'] &= [a', c', a][b, c', b'] + [c', b, a][a', c', b'] \end{aligned}$$

and this gives the assertion.

The next identity lies deeper. We challenge the reader to state it, let alone prove it in the language of Hopf algebras:

7.9. THEOREM. *Let A, B, C, X, Y be extensors, with $\text{step}(A) = k$, $\text{step}(B) = h + p$, $\text{step}(C) = q$, $\text{step}(X) = n - k - h \geq 0$, $\text{step}(Y) = n - p - q \geq 0$. Then*

$$\begin{aligned} &(A \vee X) \wedge B \wedge (C \vee Y) \\ &= (-1)^{h(n - (h + k))} \\ &\quad \times \sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2) ((A \vee B_2) \wedge X) \vee ((B_1 \vee C) \wedge Y). \end{aligned}$$

Proof. The left side can be bracketed as $((A \vee X) \wedge B) \wedge (C \vee Y)$. The leftmost meet is computed by splitting B , thereby obtaining

$$\begin{aligned} &(A \vee X) \wedge B \wedge (C \vee Y) \\ &= \left(\sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2) [A, X, B_2] B_1 \right) \wedge (C \vee Y). \end{aligned}$$

A sign computation yields $[A, X, B_2] = (-1)^{h(n - h - k)} [A, B_2, X]$, and thus the right side equals

$$(-1)^{h(n - (h + k))} \sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2) [A, B_2, X] B_1 \wedge (C \vee Y).$$

By Proposition 5.2, we have $[A, B_2, X] = (A \vee B_2) \wedge X$. Therefore, the last displayed expression equals

$$\begin{aligned} &(-1)^{h(n - (h + k))} \sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2) \\ &\quad \times (((A \vee B_2) \wedge X) B_1) \wedge (C \vee Y). \end{aligned}$$

Note that $(A \vee B_2) \wedge X$ is an extensor of step 0. Again by Proposition 5.2,

we have $B_1 \wedge (C \vee Y) = [B_1, C, Y]$ and the last displayed expression equals

$$(-1)^{h(n-(h+k))} \sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2)((A \vee B_2) \wedge X)[B_1, C, Y].$$

A third application of Proposition 5.2 to $[B_1, C, Y] = (B_1 \vee C) \wedge Y$ allows us to simplify this expression to

$$(-1)^{h(n-(h+k))} \sum_{(B_1, B_2) \in \mathcal{S}(B; p, h)} \text{sign}(B_1, B_2)((A \vee B_2) \wedge X) \vee ((B_1 \vee C) \wedge Y),$$

thereby completing the proof of the identity. ■

We shall now see that the argument of the preceding proof admits a notable generalization. The following identity, the last we shall prove in the present work, has no precedents in the literature. It can be viewed as playing a role in the double algebra similar to the role of the distributive law for union and intersection of sets in Boolean algebra. It is the crucial step in the proof of various straightening algorithms of invariant theory (see, e.g., [22] and [24]).

7.10. THEOREM. *Let $C_1, \dots, C_r$ be extensors, with $\text{step}(C_i) = n - q_i > 0$ and $q_i > 0$. Set $q_1 + \dots + q_r = p$. Let A and B be extensors, $\text{step}(A) = s$, $\text{step}(B) = k$, $s + k \geq p$. Then*

$$\begin{aligned} & (A \vee B) \wedge C_1 \wedge \dots \wedge C_r \\ &= \sum_{i_1, \dots, i_{r+1}} (-1)^{\sigma(i_1, \dots, i_{r+1})} \\ & \quad \times \sum_{(B_1, \dots, B_{r+1}) \in \mathcal{S}(B; q_1 - i_1, \dots, q_r - i_r, k + s - p - i_{r+1})} \text{sign}(B_1, \dots, B_{r+1}) \\ & \quad \times (A \wedge (B_1 \vee C_1) \wedge \dots \wedge (B_r \vee C_r)) \vee B_{r+1}, \end{aligned}$$

where the outermost sum is extended to all (ordered) $(r+1)$ -tuples of non-negative integers $(i_1, \dots, i_{r+1})$ such that $i_1 + \dots + i_{r+1} = s$, $i_j \leq q_j$ for every $j = 1, 2, \dots, r$, $i_{r+1} \leq k + s - p$ and

$$\sigma(i_1, \dots, i_{r+1}) = \sum_{j=1}^r i_{j+1} \left(\sum_{h=1}^j (q_h - i_h) \right).$$

Proof. Consider $(A \vee B) \wedge C_1 \wedge \dots \wedge C_r$ and expand the meet by splitting the extensor $A \vee B$, thereby obtaining

$$\begin{aligned} & \sum_{(D_1, \dots, D_{r+1}) \in \mathcal{S}(A \vee B; q_1, \dots, q_r, k + s - p)} \text{sign}(D_1, \dots, D_{r+1}) \\ & \quad \times [D_1, C_1] \dots [D_r, C_r] D_{r+1}. \end{aligned}$$

Let $A = a_1 \vee \cdots \vee a_s$ and $B = b_1 \vee \cdots \vee b_k$. We evaluate the above split relative to an ordered basis of which the first $s+k$ elements are $(a_1, \dots, a_s, b_1, \dots, b_k)$. Every D_i in the split can be uniquely written as

$$D_i = A_i \vee B_i.$$

Note that, in the above representation, $B_i = 1$ implies that A_i is an extensor of step q_i , for every $i = 1, 2, \dots, r$, $B_{r+1} = 1$ implies that A_{r+1} is an extensor of step $k + s - p$ and vice versa. Thus, the last display equals

$$\sum_{(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1}) \in \mathcal{S}(A \vee B; q_1, \dots, q_r, k + s - p)} \text{sign}(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1}) \\ \times [A_1 \vee B_1, C_1] \cdots [A_r \vee B_r, C_r] A_{r+1} \vee B_{r+1}.$$

We now write this as a multiple sum, namely,

$$\sum_{i_1 + \cdots + i_{r+1} = s} \sum_{(A_1, \dots, A_{r+1}) \in \mathcal{S}(A; i_1, \dots, i_{r+1})} \\ \times \sum_{(B_1, \dots, B_{r+1}) \in \mathcal{S}(B; q_1 - i_1, \dots, q_r - i_r, k + s - p - i_{r+1})} \text{sign}(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1}) \\ \times [A_1 \vee B_1, C_1] \cdots [A_r \vee B_r, C_r] A_{r+1} \vee B_{r+1};$$

the sign can be rewritten as

$$\text{sign}(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1}) = \frac{\text{sign}(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1})}{\text{sign}(A_1, \dots, A_{r+1}) \text{sign}(B_1, \dots, B_{r+1})} \\ \times \text{sign}(A_1, \dots, A_{r+1}) \text{sign}(B_1, \dots, B_{r+1});$$

we note that

$$\frac{\text{sign}(A_1 \vee B_1, \dots, A_{r+1} \vee B_{r+1})}{\text{sign}(A_1, \dots, A_{r+1}) \text{sign}(B_1, \dots, B_{r+1})}$$

depends only on the choice of $i_1, \dots, i_{r+1}$ and we can therefore write it as

$$(-1)^{\sigma(i_1, \dots, i_{r+1})},$$

where

$$\sigma(i_1, \dots, i_{r+1}) = \sum_{j=1}^r i_{j+1} \left(\sum_{h=1}^j (q_h - i_h) \right).$$

Thus, the last displayed expression simplifies to

$$\begin{aligned}
 & \sum_{i_1 + \dots + i_{r+1} = s} (-1)^{\sigma(i_1, \dots, i_{r+1})} \\
 & \times \sum_{(B_1, \dots, B_{r+1}) \in \mathcal{S}(\mathcal{B}; q_1 - i_1, \dots, q_r - i_r, k + s - p - i_{r+1})} \text{sign}(B_1, \dots, B_{r+1}) \\
 & \times \left(\sum_{(A_1, \dots, A_{r+1}) \in \mathcal{S}(\mathcal{A}; i_1, \dots, i_{r+1})} \text{sign}(A_1, \dots, A_{r+1}) \right. \\
 & \left. \times (A_1 \wedge (B_1 \vee C_1)) \cdots (A_r \wedge (B_r \vee C_r)) A_{r+1} \right) \vee B_{r+1}.
 \end{aligned}$$

The innermost sum simplifies to

$$A \wedge (B_1 \vee C_1) \wedge \cdots \wedge (B_r \vee C_r). \quad \blacksquare$$

7.11. COROLLARY. *Let $C_1, \dots, C_r$ be extensors, with $\text{step}(C_i) = n - q_i > 0$ and $q_i > 0$. Set $q_1 + \dots + q_r = p$. Let A and B extensors, $\text{step}(A) = s$, $\text{step}(B) = k$, $s + k = p$. Then*

$$\begin{aligned}
 & (A \vee B) \wedge C_1 \wedge \cdots \wedge C_r \\
 & = A \wedge \sum_{i_1, \dots, i_r} (-1)^{\sigma(i_1, \dots, i_r)} \\
 & \times \sum_{(B_1, \dots, B_r) \in \mathcal{S}(\mathcal{B}; q_1 - i_1, \dots, q_r - i_r)} \text{sign}(B_1, \dots, B_r) (B_1 \vee C_1) \wedge \cdots \wedge (B_r \vee C_r),
 \end{aligned}$$

where the outermost sum is extended to all (ordered) r -tuples of non-negative integers $(i_1, \dots, i_r)$ such that $i_1 + \dots + i_r = s$, $i_j \leq q_j$ for every $j = 1, 2, \dots, r$ and

$$\sigma(i_1, \dots, i_r) = \sum_{j=1}^r i_{j+1} \left(\sum_{h=1}^j (q_h - i_h) \right).$$

Proof. By the preceding Theorem, we have

$$\begin{aligned}
 & (A \vee B) \wedge C_1 \wedge \cdots \wedge C_r \\
 & = \sum_{i_1, \dots, i_r} (-1)^{\sigma(i_1, \dots, i_r, 0)} \\
 & \times \sum_{(B_1, \dots, B_{r+1}) \in \mathcal{S}(\mathcal{B}; q_1 - i_1, \dots, q_r - i_r, 0)} \text{sign}(B_1, \dots, B_{r+1}) \\
 & \times (A \wedge (B_1 \vee C_1) \wedge \cdots \wedge (B_r \vee C_r)) \vee 1;
 \end{aligned}$$

since the meet is bilinear, we can rewrite the above summation as

$$A \wedge \sum_{i_1, \dots, i_r} (-1)^{\sigma(i_1, \dots, i_r)} \\ \times \sum_{(B_1, \dots, B_r) \in \mathcal{S}(B; q_1 - i_1, \dots, q_r - i_r)} \text{sign}(B_1, \dots, B_r) (B_1 \vee C_1) \wedge \dots \wedge (B_r \vee C_r). \quad \blacksquare$$

8. GEOMETRIC CALCULUS

The set of all extensors (that is, we recall, of all decomposable antisymmetric tensors) of a Peano space is closed under joins and meets, and is thus an algebra in the sense of universal algebra. It will be seen presently that this algebra is a suitable setting for the statement (and, it is hoped, for the eventual solution) of the problems of classical invariant theory. To this end, we begin by developing a suitable language, aided by our previous study of the double algebra.

Let S be a set (whose elements will be called *variables*) with a grading g , namely, a function from S to the set of non-negative integers. We say that $e \in S$ is of *step* k if $g(e) = k$.

We consider the free algebra (with no identities or associativity conditions) generated by S , with two binary operations, $\vee$ and $\wedge$, which we shall call the *free join* and the *free meet*, respectively. We denote such a free algebra by $P_f(S)$, and call it the *anarchic Peano algebra* on the graded set S . Thus, an element of $P_f(S)$ is a parenthesized polynomial (in the sense of universal algebra) in the variables and in the operations of free join and free meet.

In terms of the anarchic algebra $P_f(S)$, we can define the important notion of geometric identity, as we shall now do.

Henceforth, let V be a Peano space of step n over a field K , and set $E(V)$ for the set of all extensors of V . Let S be a set with a grading g . Given a function $\phi_0: S \rightarrow E(V)$ such that $\phi_0(e)$ is of step k if $g(e) = k \leq n$ and $\phi_0(e) = 0$ otherwise, we define the notion of *canonical map* ϕ associated to ϕ_0 of the anarchic algebra $P_f(S) \circ E(V)$ as follows:

- (i) for every element $e \in S$, $\phi(e) = \phi_0(e)$;
- (ii) if $p \in P_f(S)$ and $p = p_1 \vee p_2$, set $\phi(p) = \phi(p_1) \vee \phi(p_2)$; if $p \in P_f(S)$ and $p = p_1 \wedge p_2$, set $\phi(p) = \phi(p_1) \wedge \phi(p_2)$.

This unequivocally defines $\phi(p)$ for every $p \in P_f(S)$, by induction on the number of pairs of parentheses of the polynomial p .

We say that a pair of polynomials (p, q) , with $p, q \in P_f(S)$, is a *geometric identity of dimension n over the field K* when $\phi(p) = \phi(q)$ for every canonical

map ϕ of the anarchic algebra $P_f(S)$ to the set of extensors $E(V)$ of a Peano space V of step n over the field K .

A host of classical theorems of projective geometry can be stated as geometric identities. As an example, we show that *Desargues' theorem* is a geometric identity of dimension 3 over any field, namely, the following identity.

Let a, b, c and a', b', c' be elements of step 1 in the anarchic algebra $P_f(S)$. Let ϕ_0 be any grading preserving map of S into a Peano space of step 3 and ϕ its canonical map. We identify the elements of $P_f(S)$ with their image under ϕ and write $\vee$ and $\wedge$ in place of the free join and the free meet, for ease of notation. Then the following is a geometric identity:

$$\begin{aligned} & ((b \vee c) \wedge (b' \vee c')) \vee ((c \vee a) \wedge (c' \vee a')) \vee ((a \vee b) \wedge (a' \vee b')) \\ & = (a \vee b \vee c) \wedge (a' \vee b' \vee c') \wedge ((a \vee a') \wedge (b \vee b') \vee (c \vee c')). \end{aligned}$$

In fact, in a Peano space, the left side can be written as

$$[bc \wedge b'c', ca \wedge c'a', ab \wedge a'b'] E, \quad (*)$$

where we write ab in place of $a \vee b$, and use the usual convention about bracketing. By Proposition 5.2, the above bracket can be written as

$$((bc \wedge b'c') \vee (ca \wedge c'a')) \wedge (ab \wedge a'b').$$

We now expand the meet by splitting the extensor $ab \wedge a'b'$, thereby obtaining

$$\begin{aligned} & ((bc \wedge b'c') \wedge a'b') \vee ((ca \wedge c'a') \wedge ab) \\ & - ((bc \wedge b'c') \wedge ab) \vee ((ca \wedge c'a') \wedge a'b'). \end{aligned}$$

Now, we expand $(bc \wedge b'c') \wedge a'b'$ by splitting the bracketed extensor. We obtain

$$-[b', c', a'] [b, c, b'].$$

By a similar expansion, we get

$$\begin{aligned} (ca \wedge c'a') \wedge ab &= -[c', a', a] [c, a, b] \\ (bc \wedge b'c') \wedge ab &= [b, c, a] [b', c', b] \\ (ca \wedge c'a') \wedge a'b' &= [c, a, a'] [c', a', b'] \end{aligned}$$

and then $(*)$ equals

$$[b', c', a'] [c, a, b] ([b, c, b'] [c', a', a] - [b', c', b] [c, a, a']) E. \quad (**)$$

But, by Proposition 7.1 the term in parentheses equals

$$(aa') \wedge (bb') \wedge (cc').$$

In conclusion, (**) equals

$$[a, b, c][a', b', c']((aa') \wedge (bb')) \vee cc'),$$

which is the identity to be proved.

We note that the rendering of Desargues' theorem by a geometric identity includes degenerate cases, when the vectors a, b, c or a', b', c' do not span.

A host of generalizations of Desargues' theorem obtained since the last century, and sundry other theorems of projective geometry, can be expressed as geometric identities, as we hope to show elsewhere. More important, however, is the general *problem of synthetic geometry*, a problem of noble ancestry which can now be recast in a new light. We shall see that this problem is closely related to a generalization of the second fundamental theorem of invariant theory.

Let $\text{Id}(n, K)$ be the set of all geometric identities of dimension n over the field K , namely, the set of all pairs (p, q) in $P_f(S)$ such that $\phi(p) = \phi(q)$ for every canonical map ϕ of $P_f(S)$ to the set of extensors $E(V)$, for V of dimension n over the field K . Every identity in $\text{Id}(n, K)$ defines a congruence (in the sense of universal algebra), that is, an equivalence relation which is compatible with free join and free meet. We denote by $\text{Syn}(S, n, K)$ the quotient algebra of $P_f(S)$ obtained by imposing all identities in $\text{Id}(n, K)$. Thus, $\text{Syn}(S, n, K)$ is again an algebra whose elements are equivalence classes of elements of the anarchic algebra $P_f(S)$. The two operations of $\text{Syn}(S, n, K)$ may again be written as $\vee$ and $\wedge$. We call the algebra $\text{Syn}(S, n, K)$ the *free geometric algebra of step n over the field K* , freely generated by the graded set S . Any homomorphic image of an algebra $\text{Syn}(S, n, K)$ will be called a *geometric algebra* of step n over the field K . For example, $E(V)$ is a geometric algebra, endowed with the ordinary meet and join.

Let S be graded so that every element is of degree 1. We conjecture that the free geometric algebra $\text{Syn}(S, n, K)$ can be represented as the double algebra of a Peano space of step n over a field $\bar{K}$. The field $\bar{K}$ need not coincide with the field K .

A similar conjecture can be made when the set S consists of elements of different steps. When all elements of S are either of step 1 or of step $n-1$, the conjecture is still probably valid. When, however, S includes elements of steps other than 1 or $n-1$, the analogous conjecture is not in general true, and leads to the interesting question of a representation of the free geometric algebra by a structure derived from linear algebra.

Let $D(S, K)$ be the vector space over the field K generated by the elements of the anarchic algebra $P_f(S)$ on a graded set S . The operations $\odot$ and $\oslash$ of $P_f(S)$ can be extended to the elements of $D(S, K)$ by bilinearity. An element p of $D(S, K)$ will again be called a polynomial in the variables $a_1, a_2, \dots, a_k$ of S , and written $p(a_1, a_2, \dots, a_k)$, if it is built out of these variables by joins, meets and linear combinations. If $G(V)$ is the double algebra of a Peano space V of step n over K , then again any function ϕ_0 of S to the set of extensors of $G(V)$ as above extends trivially to a homomorphism ϕ of the algebra $D(S, K)$ to the double algebra $G(V)$.

A polynomial $p(a_1, a_2, \dots, a_k)$ in $D(S, K)$ will be called an *identity of step n over the field K* when $\phi p(a_1, a_2, \dots, a_k) = 0$ for every such homomorphism ϕ . The set of all identities is a subspace of $D(S, K)$, and the quotient space $FD(S, n, K)$ will be called the *free double algebra of step n over the field K* . The algebra $FD(S, n, K)$ is again a vector space over which two bilinear operations are defined, which we again call join and meet. The structure of the free double algebra is not known. Its determination would be an extension of the second fundamental theorem of invariant theory.

Let $b_1, \dots, b_n$ be elements of step 1 in S . A polynomial $p(a_1, \dots, a_k, b_1, \dots, b_n)$ in $D(S, K)$ is said to be an invariant of the "vector" variables $a_1, \dots, a_k$ of step 1 when

$$\phi(p) = \phi'(p)$$

for every two homomorphisms which extend two maps ϕ_0 and ϕ'_0 such that $\phi_0(a_i) = \phi'_0(a_i)$ but $\phi_0(b_i) = e_i$ and $\phi'_0(b_i) = f_i$, where $e_1, \dots, e_n$ and $f_1, \dots, f_n$ are two unimodular bases. Under these conditions, the theorem of Doubilet, Rota and Stein [22] asserts that a polynomial $q(a_1, a_2, \dots, a_k)$ in $D(S, K)$ can be found, containing at most the variables $a_1, a_2, \dots, a_k \in S$, such that, denoting by Π the canonical surjection of $D(S, K)$ onto $FD(S, n, K)$, we have

$$\Pi(p(a_1, \dots, a_k, b_1, \dots, b_n)) = \Pi(q(a_1, \dots, a_k)).$$

In other words, the variables b_i can be "eliminated" when they "play no significant role," without any divisions.

Stated in these terms, the first and second fundamental theorems of invariant theory suggest similar questions for extensors of step other than 1. The notion of a free double algebra cannot be extended to describe a double algebra "freely generated" by extensors of step other than 1 or $n-1$. It can, however, be used to single out the difficulties and thereby to make a clear statement (the first, we believe) of the fundamental open problem of the invariant theory of sets of linear varieties. The problem can be formulated in terms of the notion of a *natural operation* in the free double algebra. For ease of exposition, we define the special case of a

natural operation of extensors of step j and k with extensors of step r as values, when $j, k, r \leq n$.

In the double algebra $\text{FD}(S, n, K)$ one can define the notion of an extensor in a natural way. Elements of S are naturally embedded in $\text{FD}(S, n, K)$ as such extensors, but there may be others. Let a, b, c be extensors of step j, k and r in $\text{FD}(S, n, K)$. We say that $c = g(a, b)$, that is, c is the result of applying the natural operation g on the extensors a and b , when, for every pair of homomorphisms ϕ and ϕ' of $\text{FD}(S, n, K)$ to $G(V)$ such that $\phi(a) = \phi'(a)$ and $\phi(b) = \phi'(b)$, one has $\phi(c) = \phi'(c)$.

The fundamental open problem of the invariant theory of sets of linear varieties is the problem of explicitly describing all natural operations among extensors.

We conjecture that, if the steps are fixed, a finite number of natural operations exist, in terms of which all others can be expressed. We further conjecture that all concomitants of sets of linear varieties can be expressed in terms of natural operation.

An example of such operations is provided by *plethysm*. Let $\eta_1, \dots, \eta_n$ be a unimodular basis of covectors in $G(V)$; for any antisymmetric tensor t of step k , that is, $t \in G_k(V)$, let

$$t_{i_1, \dots, i_k} = t \wedge \eta_{i_1} \wedge \dots \wedge \eta_{i_k}$$

be its Plücker coordinates. If $t^1, \dots, t^{\binom{n}{k}}$ are extensors of step k , set

$$[[t^1, t^2, \dots, t^{\binom{n}{k}}]]_k$$

to be the determinant of the matrix whose i th row consists of Plücker coordinates of t^i . We obtain in this way a Peano space on $G_k(V)$. The meet and join operations relative to this Peano space can be used to define natural operations in the free double algebra.

The open problem of higher-order syzygies for brackets, first stated by Study in [16], can be more naturally stated in the context of the free double algebra. Polynomials in $D(S, K)$ that define identities in the free double algebra $\text{FD}(S, n, K)$ form an ideal (in the sense of universal algebra), in that, if p is an identity and q is an arbitrary polynomial in $D(S, K)$, then the polynomials $p \vee q$ and $p \wedge q$ are also identities. The problem of second-order syzygies is the problem of proving that the "ideal" of identities can be generated by a finite subset, and to determine their dependencies. Higher-order syzygies can be similarly treated.

Theorem 7.10 can be made the basis of all straightening algorithms in the literature. Thus, the straightening algorithm can be seen (see [22]) as the solution of the word problem (in a special case) for the free double algebra. It is not excluded that the word problem may admit a different

solution than the one that uses the straightening algorithm, since such an algorithm does not fully exploit the powers of our main alternative law. Were all natural operations for extensors of a given step k known, one could proceed, by suitable extensions of Theorem 7.10, to extend the straightening algorithm by additional operations suggested by such natural operations.

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